Properties of Microwaves

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Abstract

Our intention is to demonstrate that electromagnetic waves exhibit the properties predicted by Malus' Law of polarisation, as well as both double slit interference and Fabry-Perot interferometry. By constructing apparata to perform each experiment, we determined a way of finding the axis along which the electric field of microwaves oscillates. Our main goal was to determine, experimentally, the wavelength of the light from our laser source. We found that the Fabry-Perot interferometry method, which involves the production of standing waves and relating them to their maxima, was a much more accurate and precise method for determining the wavelength of an unknown electromagnetic wave source. This was opposed to double-slit interference, which we found to be less precise and less accurate. The implications of these results are that light strongly exhibits wave-like characteristics when subject to these conditions. Due to the high precision of Fabry-Perot interferometry, it is highly likely that this method could be used to finely study the structure of sensitive spectral lines.

Contents

1 Introduction

Microwaves[1] are widely used for radar and wide-band communication and studying their characteristics can aid us in more efficient communication design. To gain a thorough understanding of the properties of microwaves in different contexts, we subjected a microwave source to various different scenarios and observed the intensity of the microwaves after moving through the apparatus. We looked at three different scenarios: microwave polarisation, double slit interference, and Fabry-Perot interferometry. Each setup was constructed according to the lab directions outlined by Monash University [1], and was done using a set of plates placed between the microwave source and the detector. This experiment aims to compare experimentally determined light wavelengths to the known wavelength of $\lambda = 29.46$mm.

The effect of polarisation on the microwave intensity can be predicted using Malus' Law (Equation 1).

$$I = I_0 \cos^2(\theta) \tag{1}$$

As the microwaves pass through the thin metal grating, the angle at which the grates are relative to the vertical will affect the resulting intensity [4]. In Figure 5 we see the grate is set to an angle of zero degrees, where the intensity is the highest. When the grate is vertical, the resulting intensity is predicted to be close to zero. This occurs due to the transverse nature of the orthogonal electric and magnetic waves in light. During polarisation, the electric field oscillating along the x-direction is absorbed by the grate which only allows oscillations in the y-direction. Without the electric field, the magnetic field also cannot propagate, so the intensity is much lower. As the grate rotates towards aligning with the y-direction, the electric field can pass through the grate allowing the microwaves to reach the detector. The theory predicts that there will be a gradual decrease in intensity as θ increases.

The typical double slit experiment as performed by Thomas Young in 1801 (demonstrated in Figure 7) is performed using a flat detection film a specified distance from the slits. We performed a modified version of this experiment, whereby instead of using a flat detection film, we had a single point detection device and recorded the intensity peaks and troughs as we moved the device radially. This resulted in the production of a similar phenomenon, however the angles deviated for larger angles due to the use of a curved 'surface' of detection. The governing equation behind peak location is described in Equation 2.

$$d \sin\theta = m\lambda, \text{ for } m = 0, 1, -1, 2, ..., \tag{2}$$

where m is the peak number, d is the slit separation, λ is the light wavelength, and θ is the angle the peak occurs relative to the horizontal. By varying θ and experimentally observing where the peaks occur, we can determine an experimental wavelength and compare it to the known wavelength of the microwaves produced.

Fabry-Perot interferometry is a form of multiple beam interference consisting of an internally reflected light beam that transmits some intensity on each reflection [2]. Figure 8 is an illustration of this kind of interference. Equation 3 governs the relationship between peak number and wavelength of light.

$$\frac{1}{\lambda} = \frac{m}{2d} \tag{3}$$

By experimentally determining where the peaks occur, we can use this theory to determine how accurate of a wavelength we can produce in comparison to the actual known wavelength.

2 Method

Figure 5 demonstrates the initial angle of the polarisation grate. We kept the microwave transmitter and the detection horn 50cm apart. We found that by placing the grate as close as possible to the

[1] Microwaves consist of wavelengths from 1mm to approximately 300mm

opening of the horn, we were able to minimise the effects of diffraction. However, diffraction of the light beyond the polarising grate was definitely a source of uncertainty in our data.

The oscilloscope was set up to take the average value over the past 10 divisions, where a division is a 200ms period of time. The average value is displayed as well as the largest peak to peak value measured. We took the uncertainty to be half the largest peak to peak value as displayed by the oscilloscope.

To collect the data for this experiment, we used increments of 15° for the grate. Starting at horizontal, we took a single measurement of the oscilloscope intensity (and uncertainty) at each angle increment until we reached 90°. To adjust the grate, a protractor was used and the grate was aligned to the angle on the protractor and locked in place. Since this process was done by hand, it is likely to hold some inaccuracies. In order to reach some meaningful conclusion from this data, we should expect to see a linear relationship between $\frac{I}{I_0}$ and $\cos^2(\theta)$ where the gradient is 1. To achieve this, we also recorded an I_0 value by removing the polariser entirely and taking the resulting intensity.

To perform the double slit experiment, we set up three metal screens, as pictured in Figure 6. These were set up precisely 30 ± 1cm from the receiver horn. The slit separation d is the width of the middle metal plate, measured to be 6.5 ± 0.05cm wide.

To collect data for this experiment we measured seven different data points: one at the zero angle, and three maxima/minima either side. We attached a protractor to the base of the attachment, and allowed the receiver to swivel around in a radial motion from the slits. We would expect there to be a maximum at the zero angle. We went to the next minimum either side ($m = \pm 0.5$), then to the next maximum ($m = \pm 1$), and finally the second minimum ($m = \pm 1.5$). By using Equation 2, we can create a relationship that allows us to determine a wavelength. For reference, the known microwave wavelength is $\lambda = 29.46$mm.

The final experiment was the Fabry-Perot experiment. To perform this experiment we placed two masonite[2] plates in series between the transmitter and receiver, as pictured in Figure 6. It is not significant where the plates are positioned between the transmitter and receiver, as our goal is to have a separation where a standing wave is formed between the two plates. To measure the plate separation we took a ruler measurement between the front of one plate to the front of the other plate, with a precision of ± 0.5mm. We started with a small plate separation of only a few centimetres, and made recordings at every point we observed a maximum on the digital oscilloscope. As we varied the plate separation, we made measurements of the separation every time we ran into a maximum. We determined when the maximum occurred by carefully observing the oscilloscope and recording the separation when the intensity was at its highest point after stabilising for a few seconds. Equation 3 describes a relationship between fringe number, separation distance, and wavelength. Similar to the previous experiment, we used this relation to determine an experimental wavelength for the light and compared it to the actual light wavelength.

3 Results, Analysis, and Discussion

3.1 Polarisation

During the polarisation experiment we varied the grate angle θ and observed how the polarisation of the microwaves affected the intensity received by the receiver. By plotting $\frac{I}{I_0}$ against $\cos^2(\theta)$, the resulting line gradient should be 1 to be a perfect match to Malus' Law. Our experimental data is shown in Figure 1.

A notable discussion point here is the modified form of Malus' Law used:

$$\frac{I}{I_0} = \cos^2(\theta) + c.$$

We included a constant in order to mitigate the systematic error present in our results. Outside of an ideal scenario, the value for I_0 would not be perfectly constant in time as we have assumed it to

[2] Steam-cooked and pressure-molded wood fibers

be. As a result, our data may be skewed away such that the zero point does not exactly lie on (0,0). Including this constant allows us to more accurately describe the relationship we observed, and as we will see its value is quite small in context.

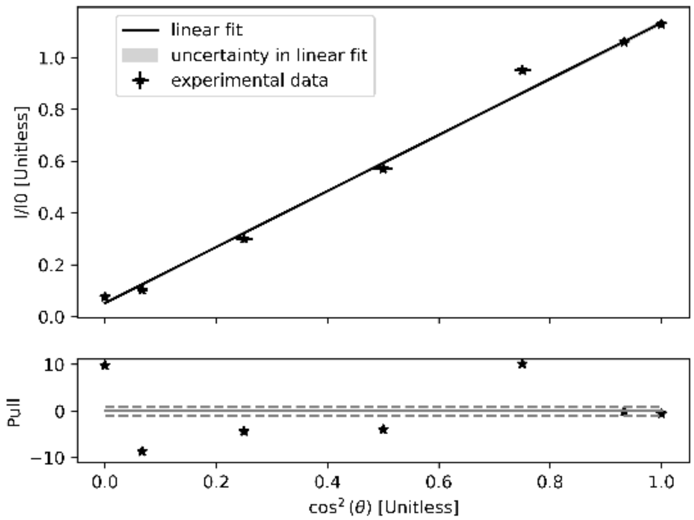


Figure 1: Polarisation data of microwaves through metal grating (top) as well as pull data (bottom) representing distance from fit line each data point is in standard uncertainties. We varied the grating angle and observed the resulting microwave intensity as observed on the digital oscilloscope. The resulting fit line closely obeys Malus' Law (Equation 1).

The resulting gradient was 1.08 ± 0.01, with a constant $c = 0.052 \pm 0.002$. The resulting p-value was extremely close to zero, allowing us to conclude that our result is statistically significant and we can reject the null hypothesis[3]. Notably, our results consistently lay greater than one standard uncertainty from the fit line. The χ^2_ν value for this plot was 61.6, indicating a very poor correlation. This does suggest that we underestimated uncertainties. One possible way to improve this would be increasing the intensity uncertainty by a set amount for each measurement, rather than simply using half the peak to peak width from the oscilloscope. Doing this would allow a greater margin of error that arises from incorrect angle measurements from the protractor, as well as the effects of diffraction affecting the receiver input.

Achieving the predicted value of 1 to within 8 standard uncertainties, given the high precision of our data analysis, is reasonably accurate.

The uncertainty in our c-value was extremely statistically significant. This suggests that in our experiment it did not correlate accurately to the zero mark, where it is predicted to be. However, as mentioned previously, due to the inconsistency in I_0 values (which we assumed constant), we would expect this to be the case. Further evidence that this assumption is the main source of error can be seen in the final two highest data points in Figure 5. They are higher than 1, which should not be possible if I_0 is the maximum (no grate) intensity of the microwaves.

3.2 Double Slit Interference

In the double slit experiment we mimicked the detection film used in the traditional experiment by radially moving the receiver horn about its axis of rotation. The most glaring difference this causes to the theory is that we may not see a perfectly uniform diffraction pattern. The curvature of the receiver's movement would cause the pattern to stretch further away from the centre. We would also

[3]Rejecting the null hypothesis means we reject the claim that the relationship we are looking for does not exist.

see a much faster drop-off of intensity with maxima occurring further from the centre. Rather than introducing a prediction theory that is marginally more complicated, involving the radius of curvature of the receiver's movement, we opted to use the same theory that is applied to a flat detector. The impact of this decision is that our final wavelength result is less accurate for the double slit experiment than it is for the Fabry-Perot experiment.

Figure 2 shows the individual data points indicating the θ points where a minimum or maximum was observed.

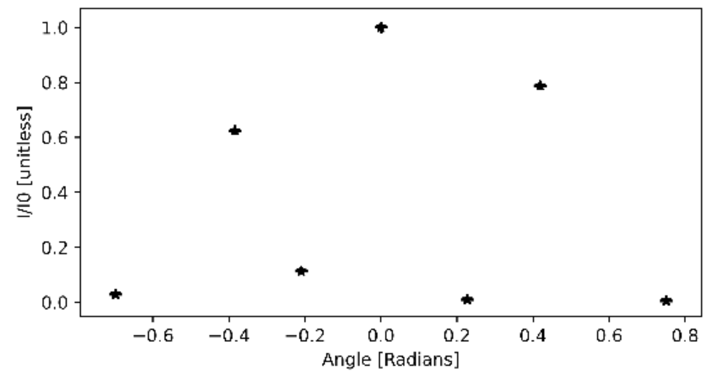


Figure 2: Raw measurement data showing the location of maxima and minima with respect to receiver angle.

Evidently, we recorded three maxima and four minima, with the central point at zero radians corresponding to a maximum. As discussed before, the resulting intensity of the two maxima either side are not as high as the central maximum. This occurs due to the detection curvature, as the interference is not as extreme and perfect as it would be if the detection plane was flat. This plot does effectively demonstrate the existence of a diffraction pattern produced as a result of our apparatus.

In extension, we may number the maxima taking $m = 0$ to be the central maximum, and use Equation 2 to determine the respective path differences $d \sin\theta$, where d is the predetermined width of the central plate. This is a useful adjustment as

$$\lambda = \frac{d \sin\theta}{m}.$$

This relationship is visualised in Figure 3.

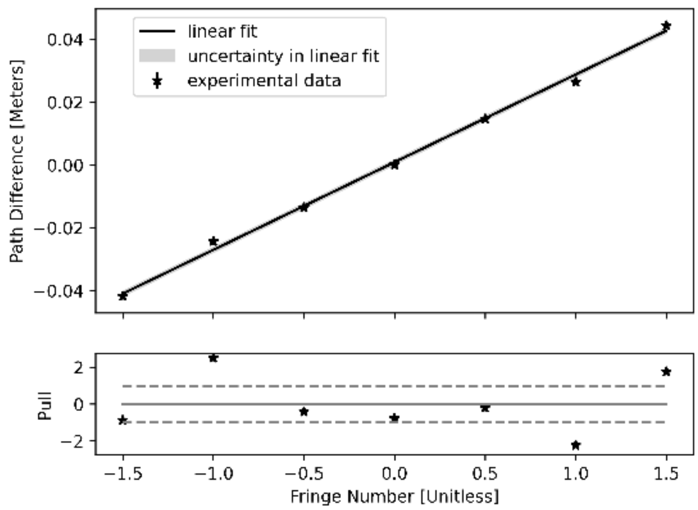


Figure 3: Path difference $d\sin\theta$ as a function of numbered fringes from Figure 2, as well as pull graph indicating deviation from fit line in standard uncertainties. By Equation 2, the gradient of this plot is the wavelength.

The p-value of this plot is $7.11 \cdot 10^{-3}$, suggesting a high statistical significance. We may reject the null hypothesis and conclude that there is a significant relationship between fringe number and path difference. We may also observe that close to two thirds of our data points lie within one standard uncertainty of the fit line, indicating further that the correlation is likely to be observed to a reasonably high degree of precision. More quantitatively, the χ^2_ν value for this plot was 3.18, indicating that the data and fitline match in accord with the measurement error.

As discussed above, the gradient of this plot represents the experimental wavelength of light for this experiment. This was calculated to be

$$\lambda_{\text{pred}} = 27.9 \pm 0.4\text{mm}.$$

This lies within 4 standard uncertainties of the actual wavelength of 29.46mm. Such a result suggests a high probability that the prediction based Equation 2 was observed in this experiment.

Another possible source of uncertainty with this experiment may arise from our measurement of the angle θ. We measured this based on a protractor attached to the bottom of the apparatus, and we estimated a $1°$ uncertainty in this measurement. In reality this may have been higher as we only relied on one lab partner to perform this measurement. A better way to do this would be to have multiple people make independent measurements and take an average of the value.

3.3 Fabry-Perot Interferometry

The final experiment we performed was the Fabry-Perot Interferometry experiment. Our goal was to position the two masonite plates at specific distances apart such that standing waves would be formed between them. The unique property of masonite is that it allows for wave transmission and reflection simultaneously, as shown in two dimensions in Figure 8. In our case, however, this is occurring in one dimension. The full Fabry-Perot relation involves the angles of the incoming and outgoing rays, as well as the refractive index within the material and outside the material. In our case, the angles are all zero (one dimension) and the refractive index is only a negligible variation from 1. We also neglect any thickness of the masonite plates and, thus, any possible refraction that might occur as a result of it.

The set of maxima of successive harmonics and their respective plate distances is plotted in Figure 4. Keep in mind that the separation had an uncertainty of ± 0.05cm.

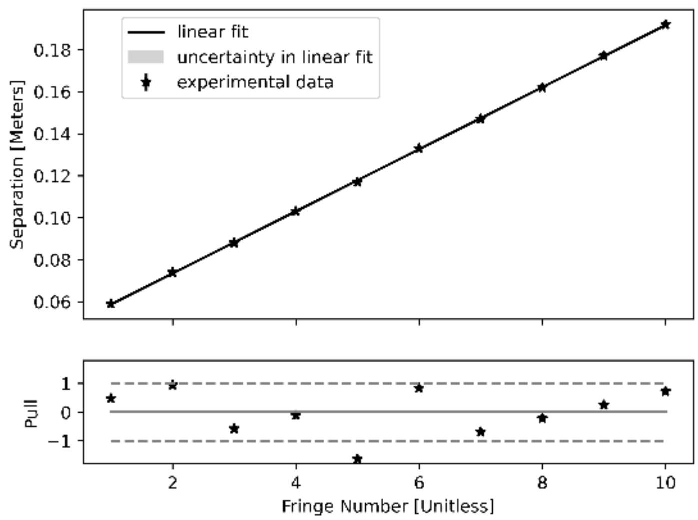


Figure 4: Plot of the Fabry-Perot interferometry data (top) where the plate separation is recorded in correspondence with the occurrence of intensity maxima. Pictured below is the pull graph for the data in standard uncertainties. By Equation 3, the gradient is half the wavelength.

We observe an extremely strong correlation between the occurrence of maxima at specific plate separations. The majority of our data points lie within one standard uncertainty of the fit line, indicating a highly precise result. The χ^2_ν value for this plot is 0.75, suggesting an extremely high correlation between the data and the fit line. This may also suggest the data has been slightly overfitted, which may arise from overestimated uncertainties. Nonetheless, this is very strong evidence to suggest the theory has been observed in this experiment.

From Equation 3, we can form the relationship

$$\lambda = 2 \cdot \frac{d}{m},$$

meaning the experimental wavelength will be given by twice the gradient of this graph. Performing this calculation,

$$\lambda_{\text{pred}} = 29.5 \pm 0.1\text{mm},$$

which is less than one standard uncertainty from the actual wavelength value of 29.5mm. This tells us that the Fabry-Perot Interferometry method was an extremely accurate experiment allowing us to confidently determine a precise and accurate result for the wavelength.

The Fabry-Perot interferometry method for calculating λ is superior to the double slit method as it is both more accurate and more precise.

The double slit interference method has an accuracy percentage of 5.25% to the actual value, while the interferometry method is 0.25% accurate. In addition, in terms of uncertainties, the interference method is 1.3% precise while the interferometry method is 0.37% precise. This is based off path length $d \sin\theta$ and plate separation respectively.

4 Conclusion

Through our exploration of the properties of electromagnetic microwaves when subject to different conditions, we determined some interesting results that ultimately confirmed some of the theory we know about light.

We were able to determine that the orientation of the electric field of our microwaves aligned with the vertical axis, which we determined by varying the angle along which a polarising grate was situated between a microwave transmitter and receiver. The prediction follows that Malus' Law will determine the change in intensity resulting from light polarisation (Equation 1). We found that our results had a high correlation with the line predicted by Malus' Law, with our gradient of 1.08 lying within 8 standard uncertainties of the predicted gradient of 1. It is notable that we underestimated our uncertainties for this experiment, resulting in highly precise values. Given this, the result is reasonably accurate.

We also performed two experiments with the microwaves to see if we could experimentally determine the wavelength of the light, had we not previously known it. Given our useful hindsight, we know the value of the light wavelength to be 29.5mm.

The first experiment we used to achieve this was the double slit interference experiment. By measuring where maxima occurred along a detection radius a specified length from the slits, we observed a distinct diffraction pattern. Using the interference law described by Equation 2, we determined a predicted wavelength of 27.9 ± 0.4mm, which is within 4 standard uncertainties of the known value.

The second experiment used Fabry-Perot interferometry to determine the wavelength, using a series of plates to form harmonics of standing waves. By recording peaks in transmittion intensity at specific plate separations, the relation described by Equation 3 allowed us to determine a wavelength. We predicted the wavelength using this method to be 29.5 ± 0.1mm, within one standard uncertainty of the known value.

Evidently, the interferometry experiment gave a much more accurate and more precise result, allowing us to conclude that it is a better method for finding the wavelength of light.

If future experiments were performed based on this theory, it would be beneficial to repeat the experiments multiple times and take average results for each run-through. This would help mitigate the occurrence of outliers and random error. In order to mitigate the potential systematic error in the experiments, it would be beneficial to use a different detection system in the double slit experiment as opposed to a swivelling receiver. A flat planar detection surface would produce a much more distinct diffraction pattern without as strong of an intensity dropoff between maxima.

References

[1] Findlay, Scott: *EM1 Introduction to Microwaves*. 2024. – `https://learning.monash.edu/mod/book/view.php?id=1903319&chapterid=298317` Accessed: 20/10/2024

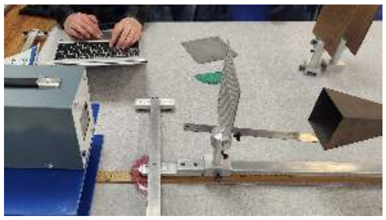
[2] Hariharan, P.: *Multiple Beam Interference*. 1992. – `https://www.sciencedirect.com/science/article/pii/B9780080918617500103` Accessed: 20/10/2024

[3] OpenStax: *27.3 Young's Double Slit Experiment*. 2019. – `https://jwu.pressbooks.pub/collegephysics/chapter/youngs-double-slit-experiment/` Accessed: 20/10/2024

[4] Ready, Science: *Polarisation of Light*. 2024. – `https://scienceready.com.au/pages/polarisation` Accessed: 20/10/2024

5 Appendix

5.1 Apparata

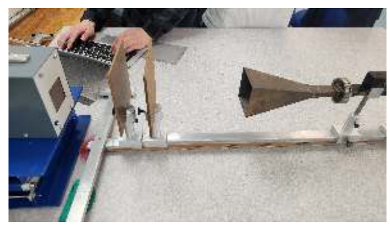


(a) Polarisation apparatus showing metal grate placed between microwave source and detector.

(b) Digital oscilloscope used for intensity measurements.

Figure 5: Apparatus pictured for polarisation component, as well as oscilloscope.

(a) Plate setup to mimic Young's double slit experiment.



(b) Two plates in series to perform Fabry-Perot interferometry, producing a standing wave between the plates.

Figure 6: Apparata for the double slit experiment and the Fabry-Perot experiment.

5.2 Diagrams and Examples

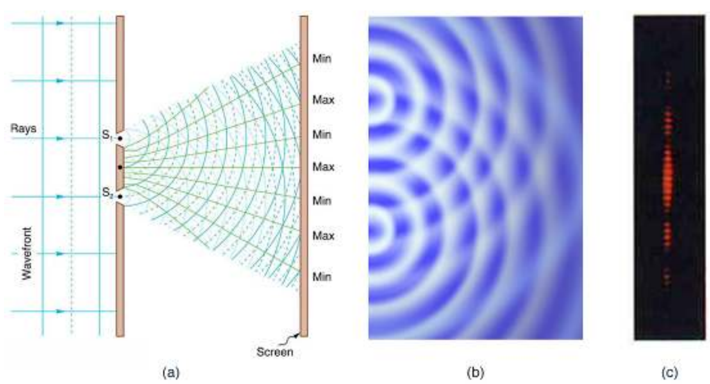


Figure 7: A typical example of Young's double slit experiment which results in maxima and minima occurring along a flat detection film in two dimensions [3].

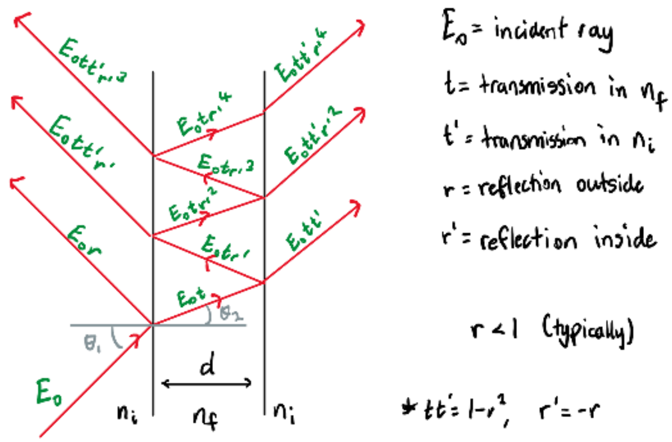


Figure 8: A diagram of multiple beam interference seen in Fabry-Perot interferometry. Keep in mind that in our experiment we did this along only a single dimension, not as shown.

5.3 Raw Data

Angle (°)	u_Angle (°)	Intensity (mV)	u_I (mV)
0	1	463	3
15	1	435	3
30	1	390	3
45	1	234	2
60	1	123	2
75	1	42	1
90	1	31	1

Table 1: Raw data values for the polarisation experiment.

Angle (°)	u_Angle (°)	Maximum	Intensity (mV)	u_I (mV)
0	1	0	213	3
13	1	0.5	2	2
-12	1	-0.5	24	2
-22	1	-1	133	3
24	1	1	168	2
43	1	1.5	1	1
-40	1	-1.5	6	2

Table 2: Raw data values for the double slit experiment.

Peak intensity (mV)	**u_I (mV)**	**Maximum**	**Distance (cm)**	**u_Distance (cm)**
702	2	1	5.9	0.05
699	3	2	7.4	0.05
700	2	3	8.8	0.05
701	3	4	10.3	0.05
704	3	5	11.7	0.05
704	2	6	13.3	0.05
714	2	7	14.7	0.05
721	2	8	16.2	0.05
723	2	9	17.7	0.05
726	2	10	19.2	0.05

Table 3: Raw data values for the Fabry-Perot Interferometry experiment.